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Inventor(s)

Gardinier; Stephane et al.

SYSTEM AND METHOD FOR PLANT AGRICULTURE

Abstract

The invention involves a system and method for plant agriculture. The system includes a sheet of barrier material upon which soil is placed. A drip tube is placed in the soil so that the soil can be shaped, and the sheet material is folded over the top of the soil so that the edges of the sheet overlap. A seal, which may be soil, is placed on the overlapping edges of the sheet to hold the sheet material in a closed position. The crop is planted through the sheet material, and growth occurs through the aperture created during planting.

Inventors: Gardinier; Stephane (Sarasota, FL), Jones; Lee C. (Okeechobee, FL), McManus; Matthew (Sarasota, FL)

Applicant: Jardin Holdings I, LLC (Sarasota, FL)

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Background/Summary

PRIORITY CLAIM [0001] In accordance with 37 C.F.R. 1.76, a claim of priority is included in an Application Data Sheet filed concurrently herewith. Accordingly, the present invention claims priority as a divisional application to U.S. application Ser. No. 16/697,901, entitled “SYSTEM AND METHOD FOR PLANT AGRICULTURE”, filed Nov. 27, 2019, which claims priority to U.S. Provisional Patent Application No. 62/772,975, entitled “SYSTEM AND METHOD FOR PLANT AGRICULTURE”, filed Nov. 29, 2018. The contents of the above referenced applications are incorporated herein by reference in their entirety.

FIELD OF INVENTION

[0002] The present invention generally relates to agriculture and, more particularly, to a plant agriculture system and method of growing plants to maximize yield while reducing water usage and chemical runoff.

BACKGROUND INFORMATION

[0003] Plant agriculture is defined as any plant, or part thereof, grown, maintained, or otherwise produced for commercial purposes, including growing, maintaining, or otherwise producing plants for sale or trade, for research or experimental purposes, or for use in part or in their entirety in another location. Plant agriculture is always concerned with improving production and reducing any harmful effects resulting from use of the land to grow plants.

[0004] The water in brooks, streams and creeks around the world carry a heavy load of nitrates from fertilizers used for plant agriculture. These nitrates eventually make their way downstream to rivers, lakes and oceans, carrying the nitrogen and oxygen from the nitrates and fertilizing blooms of algae that deplete oxygen in the water, often endangering fish and wildlife. Typically, bacteria in the water remove the excess fertilizer from the water through a chemical process known as denitrification, which enables the bacteria to convert the nitrate to nitrogen that is released to the atmosphere as a gas.

[0005] Over-fertilization is a leading cause of fertilizer (nitrate) runoff. Over-fertilization occurs when more nutrients are added to the soil than the plants can utilize; unused nutrients are then carried away by irrigation or rain water. Several factors can result in over-fertilization. It can be difficult to determine the nutrient content of some organic fertilizers; as a result, excess nutrients may exceed the estimated content. Farmers may also misdiagnose poor plant growth as a nutrient deficiency, and add fertilizer as a corrective measure.

[0006] Improper irrigation and control of water runoff is another cause of problems related to fertilizer runoff. Unfortunately, water control from large farm fields is nearly impossible due to weather, which may drop large amounts of water in a short period. The water follows the natural contours of the land to reach ditches, streams and creeks which flow further downstream to the larger bodies of water until flow reaches the ocean.

[0007] Potential solutions to the problems are difficult to implement. We could quit growing crops such as corn, which is used for fuel and feed for animals. Alternatively, we could quit fertilizing fields and treating them with herbicides and pesticides. However, neither of these approaches is practical or realistic given the amount of food that must be produced per acre of land to feed the people of the world.

[0008] Therefore, what is needed is a system and method of farming that not only reduces runoff but also reduces the need for water, pesticides and herbicides, and will produce at least as much food per acre as the present methods. The system should be relatively easy to implement in order to achieve acceptance by the end user. The system must be easily and quickly assembled using

minimal equipment and requiring a minimal number of persons.

[0009] Thus, the present invention provides a system and method of plant agriculture which overcomes the disadvantages of prior art systems. The system and method of plant agriculture of the present invention not only provides for relative ease in the assembly and use, it also permits reduction in the amount of water necessary for proper plant production. The present invention also reduces the amount of herbicides and pesticides needed, thereby reducing the amount of undesired runoff of chemicals that causes problems with algae blooms and the like.

SUMMARY OF THE INVENTION

[0010] Briefly, the invention involves a system and method for plant agriculture. The system includes a sheet of barrier material upon which soil is placed. A drip tube is placed in the soil so that the soil can be shaped, and the sheet material is folded over the top of the soil so that the edges of the sheet overlap. A seal, which may be soil, is placed on the overlapping edges of the sheet to hold the sheet material in a closed position. The crop is planted through the sheet material, and growth occurs through the aperture created during planting.

[0011] Accordingly, it is an objective of the present invention to provide a system and method of plant agriculture.

[0012] It is a further objective of the present invention to provide a system and method of plant agriculture that reduces water requirements for plant growth.

[0013] It is yet a further objective of the present invention to provide a system and method of plant agriculture that reduces runoff of water, including chemicals and leaching of chemicals into soil proximate plant growth.

[0014] It is another objective of the present invention to provide a plant agriculture system that reduces the need for chemical application to the soil by application of the chemicals to a sheet material which provides a barrier to adjacent soil.

[0015] Other objectives and advantages of this invention will become apparent from the following description taken in conjunction with the accompanying drawings wherein are set forth, by way of illustration and example, certain embodiments of this invention. The drawings constitute a part of this specification, include exemplary embodiments of the present invention, and illustrate various objects and features thereof.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0016] FIG. 1 is a partial top perspective view of the prior art to the present invention;

[0017] FIG. 2 is a partial top perspective view, illustrating one embodiment of field preparation for installation of the present system;

[0018] FIG. 3 is a partial top perspective view of the embodiment shown in FIG. 2, illustrating assembly of the present system;

[0019] FIG. 4A is a partial top perspective view of the embodiment shown in FIGS. 2 and 3, illustrating the present system ready for planting;

[0020] FIG. 4B is a partial top perspective view of the embodiment shown in FIGS. 2 and 3, illustrating the present system ready for planting and having the soil formed into a trapezoid shape when viewed from the end;

[0021] FIG. 4C is a partial top perspective view of the embodiment shown in FIGS. 2 and 3, illustrating the present system ready for planting and having the soil formed into a rectangle shape when viewed from the end;

[0022] FIG. 5 is a pictorial view, illustrating plant growth using the present system and method;

[0023] FIG. 6 is a graph illustrating total biomass growth of tomatoes using the present system;

[0024] FIG. 7 is a graph comparing first harvest yields of the present system compared to the prior

art system; and

[0025] FIG. **8** is a graph illustrating the water use requirements of the present system compared to the prior art systems.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0026] While the present invention is susceptible of embodiment in various forms, there is shown in the drawings and will hereinafter be described a presently preferred embodiment with the understanding that the present disclosure is to be considered an exemplification of the invention and is not intended to limit the invention to the specific embodiments illustrated.

[0027] Referring generally to FIG. **1**, a prior art system of plant growth **10** of produce, such as tomatoes or peppers, is illustrated. The plant growth system **10** includes a sheet of plastic **12** covering an elongated mound of soil **14**. A second mound of soil **16** is piled onto the edges **18** of the plastic sheet **12** to hold the plastic **12** in position. A drip tube **20** is placed in the soil to provide water to the growing plants. While this system cuts down on weed growth and reduces evaporation to the atmosphere, it fails to control leaching of water, herbicides and pesticides to the soil below.

[0028] Referring to FIGS. **2-4**, one embodiment of the plant agriculture system and method **100** is illustrated. The plant agriculture system **100** of the preferred embodiment includes ground preparation, which may include forming a depression **22** in the topsoil **24**. The depression **22** is relatively shallow and may include sidewalls **26**. The depression **22** is generally sized to allow a suitable amount of topsoil **24** to be moved to a side position adjacent the depression **22**, as illustrated in FIG. **2**. A piece of sheet material **28** is laid along the depression **22** with a remainder **30** of the sheet material **28** positioned on one side of the depression **22** and a distal edge **32** extending along the opposite side of the depression **22**, thereby covering the depression. The depression topsoil **24** from the side position is moved onto the sheet material **28** with the distal edge **32** still protruding from the depression **22**, while the remainder **30** of the sheet material **28** extends adjacent the depression. A drip tube **34** is positioned within and extends along the length of the topsoil **24**. The topsoil **24** may be formed into a desired shape with rollers, blades and the like. Suitable shapes include, but should not be limited to, a trapezoid or a rectangle when viewed from the end. Once the topsoil **24** is shaped, the remainder **30** of the sheet material **28** is positioned across the top surface of the topsoil **24** having the distal edges **32** positioned in an overlapping fashion. In the preferred embodiment, a small portion of the topsoil **24** is positioned on top of the overlapping distal edges **32** to hold the sheet material **28** in a closed position as illustrated in the middle illustration of FIG. **4**. Alternatively, the distal edges **32** may be glued, welded, stitched **36**, or otherwise secured together to form an elongate tube **38** of the sheet material **28** containing the topsoil **24**, as illustrated on the left portion of FIG. **4**. Once the elongate tube **38** is formed, seeds, seedlings, plants or the like may be planted through the sheet material **28** as is known in the art, so that the roots are positioned within the elongate tube and the remainder of the plant extends out of the elongate tube. While the figures illustrate the distal edges **32** of the sheet material **28** as being positioned on one side of the depression **22**, it should be noted that the distal edges **32** may be positioned at any point around the topsoil **24**, so long as the distal edges **32** are secured together to form the elongated tube **38** to hold in the water and fertilizer, without departing from the scope of the invention. The system **100** is also extremely versatile for a variety of crops and crop types. For example, the depression can be sized to create beds of topsoil of various sizes for different crops. As a non-limiting example, for strawberries and tomatoes, a trapezoid or rectangle of topsoil **24** could be sized for eighteen inches of depth and twenty-four inches of width. Watermelons may need six to twelve inches of topsoil depth and thirty-six inches of width. The elongated tubes **38** of topsoil **24** may be spaced to have longitudinal centerlines that are at any suitable distance apart with respect to each other for equipment to pass between the tubes. As a non-limiting example, six feet apart is very common for watermelon growth and harvest. However, tubes for strawberries may be placed closer together provided the equipment will fit between or over the top of the tubes. In this manner, the present system **100** provides great versatility for farming techniques that are

employed all over the world for growing various crops. In places where farm land is scarce, the longitudinal centerlines of the crops may be moved closer together and the crops harvested by hand to increase the density of production on smaller tracts of land while minimizing the need for fertilizer and water. The present system also provides a method and system **100** for farming areas where the climate is generally thought of as being too arid to successfully grow crops. Because the water is contained within the sheet material **28**, these areas may now be utilized to produce food for an ever increasing population.

[0029] Still referring to FIGS. **2-4**, the sheet material **28** is preferably a polymer that may be biodegradable. In some embodiments, the sheet material **28** is plastic sheeting of a suitable thickness to withstand weathering for at least one growing season, and may be constructed to withstand two or more growing seasons; whereby at least two sets of plant crops are grown in the elongate tubes **38** before they must be deconstructed and reconstructed again. In at least some embodiments, the sheet material **28** is coated on one side. The coating **39** may include, but should not be limited to, fertilizer(s), herbicides, pesticides and suitable combinations thereof. In this manner, the amounts of chemicals applied to the topsoil **24** are closely controlled, while leaching and run off are minimized.

[0030] Referring to FIG. **5**, seedlings **40**, as well as full grown plants **42**, are illustrated. As shown, conventional methods of staking **44** and tying **46** plants, such as tomato plants, are useful with the elongated tubes **38**, and prevent the stakes **45** from puncturing the tubes **38**. The plant agriculture system and method **100** is suitable for use on typical farm fields that may range from a few acres to several hundred acres without departing from the scope of the invention.

[0031] Referring to FIGS. **6-8**, various graphs are illustrated. The information to produce the graphs was gathered through a study at a Florida University. The information gathered illustrates that the present plant agriculture system **100** increases growth of the plant biomass **48** at all stages of the plant growth. Increased biomass typically results in increased production of edible product by the plants. This particular study utilized tomato plants. Referring to FIG. **7**, the study indicated that the tomato yield **50**, using the present system, resulted in an increase of 20% when compared to the prior art systems. The study also indicates that the amount of extra-large produce was significantly increased when compared to the prior art. Still yet, the study shows, in FIG. **8**, that water usage by the plants was reduced by 62% when compared to the prior art systems. Thus, the present system **100** has been shown to significantly outperform the prior art growing systems.

[0032] It is to be understood that while a certain form of the invention is illustrated, it is not to be limited to the specific form or arrangement of parts herein described and shown. It will be apparent to those skilled in the art that various changes may be made without departing from the scope of the invention, and the invention is not to be considered limited to what is shown and described in the specification.

[0033] One skilled in the art will readily appreciate that the present invention is well adapted to carry out the objects and obtain the ends and advantages mentioned, as well as those inherent therein. Any compounds, methods, procedures and techniques described herein are presently representative of the preferred embodiments, are intended to be exemplary, and are not intended as limitations on the scope. Changes therein and other uses will occur to those skilled in the art which are encompassed within the spirit of the invention and are defined by the scope of the appended claims. Although the invention has been described in connection with specific preferred embodiments, it should be understood that the invention as claimed should not be unduly limited to such specific embodiments. Indeed, various modifications of the described modes for carrying out the invention which are obvious to those skilled in the art are intended to be within the scope of the following claims.

Claims

- 1.** A method of plant agriculture comprising: Forming an elongated depression (22) by removing a portion of the topsoil (24) from a farm field, said depression (22) being sized to provide a sufficient amount of the topsoil to construct a shaped crop bed for growing an agriculture crop to harvest, the depression being formed shallow and including sidewalls (26), moving said removed topsoil (24) to a side position adjacent said depression (22), laying a piece of sheet material (28) along the length of said depression with a remainder (30) of said sheet material (28) positioned on one side of said depression (22) and a distal edge (32) of said sheet material (28) extending along an opposite side of said depression (22), thereby covering said depression (22), moving said removed topsoil (24) from the side position adjacent said depression (22) onto said sheet material (28) with said distal edge (32) of said sheet material (28) still protruding from said depression (22), while said remainder (30) of said sheet material (28) extends adjacent said depression, positioning a drip tube (34) along the length of said removed topsoil (24), forming said removed topsoil (24) into a desired shape constructing the crop bed, positioning said remainder (30) of said sheet material (28) including a second distal edge across a top surface of said removed and shaped topsoil (24) having the distal edges (32) of said sheet material (28) positioned in an overlapping position with respect to each other, said distal edges (32) secured together to hold said sheet material (28) together in a closed position.
 - 2.** The method of claim 1 wherein said distal edges (32) are secured together by placing a portion of said removed topsoil (24) on a top surface of said overlapping distal edges (32).
 - 3.** The method of claim 1 wherein said distal edges (32) are secured together by gluing said overlapping distal edges (32) together.
 - 4.** The method of claim 1 wherein said distal edges (32) are secured together by heat welding said overlapping distal edges (32) together.
 - 5.** The method of claim 4 wherein said heat welds are intermittent, having spaces of unwelded areas between said welds.
 - 6.** The method of claim 1 wherein said distal edges (32) are secured together by stitching (36) said overlapping distal edges (32) together.
 - 7.** The method of claim 1 further including planting seeds within said removed topsoil (24) through said sheet material (28) positioned across said top surface of said removed and shaped topsoil (24).
 - 8.** The method of claim 1 further including planting seedlings (40) within said removed topsoil (24) through said sheet material (28) positioned across said top surface of said removed and shaped topsoil (24).
 - 9.** The method of claim 1 including planting plants (42) within said removed topsoil (24) through said sheet material (28) positioned across said top surface of said removed and shaped topsoil (24) so that the roots of said plants (42) are positioned within said removed topsoil (24) and a remainder of said plants (42) extends out through said sheet material (28).
 - 10.** The method of claim 1 wherein said removed topsoil (24) is formed into a trapezoid shape on top of said sheet material (28).
 - 11.** The method of claim 1 wherein said removed topsoil (24) is formed into a rectangle shape on top of said sheet material (28).
 - 12.** The method of claim 1 wherein said sheet material (28) includes a coating (39) on at least one surface thereof.
 - 13.** The method of claim 12 wherein said coating (39) is a fertilizer.
 - 14.** The method of claim 12 wherein said coating (39) is an herbicide or a pesticide.
 - 15.** The plant agriculture system (100) formed by the method of any preceding claim.
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